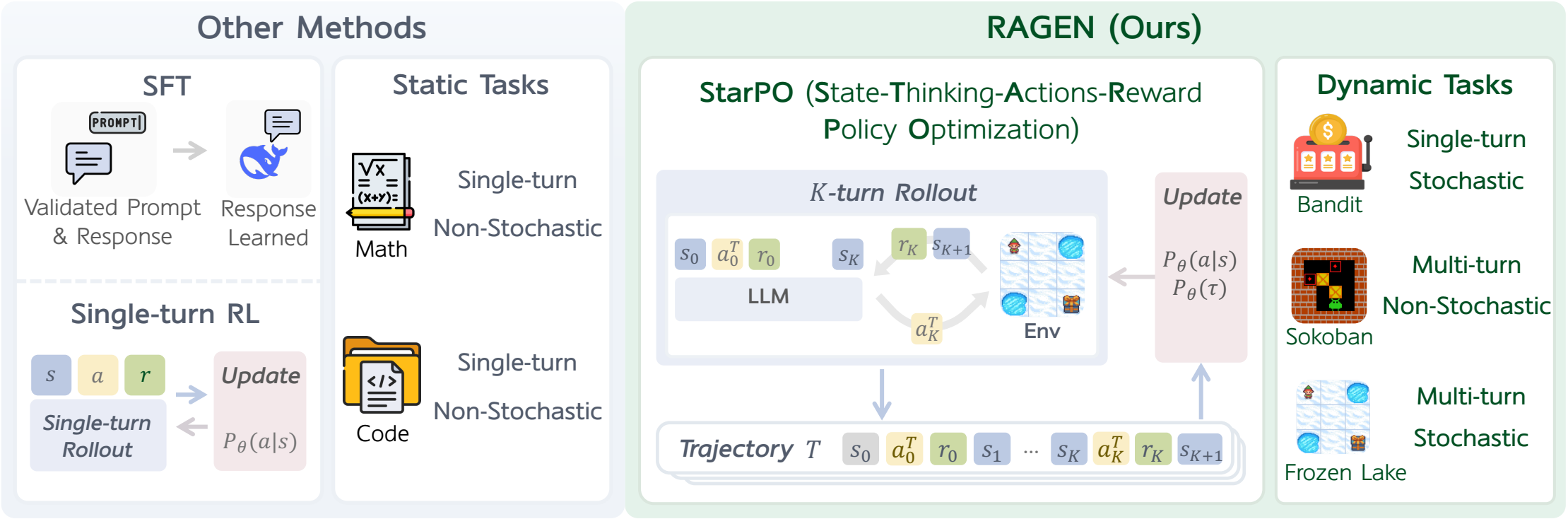

Other Methods
SFT
PROMPT
Validated Prompt & Response
Response Learned
Single-turn RL
s
a
r
Single-turn Rollout
Update
$P_\theta(a|s)$
Static Tasks
Math
Single-turn
Non-Stochastic
Code
Single-turn
Non-Stochastic
RAGEN (Ours)
StarPO (State-Thinking-Actions-Reward Policy Optimization)
K-turn Rollout
s_0 a_0^T r_0
s_K
r_K s_{K+1}
LLM
a_K^T
Env
Update
$P_\theta(a|s)$
$P_\theta(\tau)$
Trajectory T
s_0 a_0^T r_0 s_1 ... s_K a_K^T r_K s_{K+1}
Dynamic Tasks
Bandit
Single-turn
Stochastic
Sokoban
Multi-turn
Non-Stochastic
Frozen Lake
Multi-turn
Stochastic